

Source-Traceable Research Intelligence: Evidence-Grounded Discovery, Question Answering, and Thesis-Extension Support

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Abstract—Publication lists expose outputs but do not answer full-text questions or help students formulate bounded, evidence-linked projects. We report an artefact-oriented engineering case study and controlled offline evaluation of a source-traceable research-intelligence platform. From 134 catalogue records and 98 local PDFs, deterministic eligibility rules admitted 96 papers and 719 page-aware chunks. On a 50-case AI-reviewed silver set, held-out keyword retrieval achieved Recall@10 1.000 and MRR 0.947; a development-tuned hybrid achieved 0.925 and 0.860, with no Holm-corrected evidence of superiority. Review of 50 extractive answers labelled 398/400 claims supported, 2 partial, and 0 unsupported, yet citation correctness was 0.625, answer-point coverage 0.136, and all four unanswerable cases were answered. A 28-profile AI-proxy comparison estimated a finder relevance change of 0.024 (95% CI [-0.024, 0.071]). Thus traceability made weak citation targeting, completeness, abstention, and recommendation evidence visible; it did not establish human advisory benefit. The contribution is a reproducible platform and evaluation protocol that separates source facts, generated suggestions, empirical quality, and review state.

Index Terms—research intelligence, scholarly search, retrieval-augmented generation, recommendation, source traceability, reproducibility

I. INTRODUCTION

Large scholarly indexes connect papers and people [1]–[3], while recommender and expert-finding research supports content/profile matching [4], [5]. A laboratory archive has a narrower but less forgiving problem: missing PDFs, uncertain identities, generated advice, and incomplete review must remain visible instead of being hidden behind a polished search box.

The TTLAB precursor collected and displayed publications, generated lay summaries and notifications, and supported manual podcast preparation; evaluated full-text RAG was future work [6]. We extend it with full-paper ingestion, page-aware retrieval, citation-returning question answering, thesis-extension suggestions, topic/author exploration, and correction records. Here *source-traceable* means that outputs retain paper, chunk, page/section, provider, timestamp, warning, and review-state locators. It does not mean that a claim is entailed, complete, novel, feasible, or endorsed.

Three questions organize the study: **RQ1**, can a heterogeneous archive become an internally consistent corpus and how do five retrieval modes compare; **RQ2**, how strong are evidence-linked answers, extension suggestions, topics, and author links under their stated evaluation constructs; and **RQ3**, how reproducible, reviewable, and responsibly bounded is the resulting artefact? Contributions are a hash-bound corpus/index contract; a modular evidence-first platform; executed retrieval, QA, recommendation, topic, section, performance, and review evaluations; and a sanitized reproduction boundary. No recruited participants were used, so usefulness and supervisor suitability remain untested.

II. RELATED WORK

Scholarly discovery systems progressed from autonomous citation indexing to literature/author graphs and open entity catalogues [1]–[3]. Their scale is useful for discovery, but a small institutional system must also expose why a paper was included, which file supplied a passage, whether an identity is confirmed, and whether generated material has been reviewed. Research-paper recommendation surveys likewise show that methods, user models, and evaluation protocols vary widely [4]; an evidence match is not the same as a feasible supervisory recommendation. Expert-finding models infer expertise from documents [5], so our author links are deliberately phrased as publication-derived associations rather than availability or endorsement.

BM25-style lexical ranking remains a strong baseline [7]. Feature hashing is a memory-bounded sparse representation, not a learned semantic encoder [8]. Sentence-BERT and DPR provide learned sentence/passage representations [9], [10]; BEIR cautions that rankings vary across tasks and domains [11]. We therefore compare modes rather than presuming dense or hybrid superiority.

RAG conditions generation on retrieved evidence [12], and later work adds retrieval/self-critique controls [13]. Evaluation frameworks inspect context relevance and faithfulness [14], [15]; atomic decomposition exposes claim-level support [16]. Returned citations nevertheless can target the wrong passage, and high support can coexist with missing answer points. Topic

TABLE I
FROZEN CORPUS AND INDEX BOUNDARY.

Catalogue	134	Local PDFs	98
Eligible papers	96	Needs text/review	36
Raw chunks	735	Eligible chunks	719
PDF mismatches	2	Unknown sections	199

interpretability likewise requires inspection rather than relying on aggregate coherence [17]. Human-in-the-loop guidance motivates visible uncertainty and correctable outputs [18].

III. METHODOLOGY AND SYSTEM

A. Units, Corpus, and Evidence Boundary

The study combines design-science-oriented artefact construction with controlled offline evaluation. Units are papers/chunks for corpus integrity; queries for retrieval; atomic claims, citation links, and expected answer points for QA; synthetic profiles and ranked paper sets for recommendation; papers/labels for topics; and executable stages for performance. TTLAB authorized the project and corpus use, but that is neither ethics-board approval nor blanket third-party redistribution permission.

Design decisions were mapped to observable failures. Deterministic import and hash-checked indexes address silent corpus drift; page locators and typed evidence address unverifiable generation; evidence-only Finder output addresses users who do not want generated advice; append-only review events address untraceable correction; and explicit unavailable states address missing providers or PDFs. Each mechanism is evaluated at the narrowest valid unit instead of treating a passing user-interface test as retrieval evidence or a citation count as faithfulness.

The frozen snapshot named in the manifest starts with 134 records and 98 available PDFs. A paper enters experiments only when metadata, a permitted PDF, extracted text, and title/content checks agree. Table I shows the resulting boundary: 96 papers and 719 of 735 chunks are eligible; 36 records need text/review, 2 PDFs are mismatches, and 199 eligible chunks retain an Unknown section instead of a forced label. A 40-case AI-silver check measured 0.900 section accuracy.

Keyword FTS, 256-dimensional signed feature hashing, and the pinned 384-dimensional `all-MiniLM-L6-v2` encoder share the same ordered eligible-chunk boundary. Manifests bind corpus/configuration hashes, model revision, dimensions, code, ordered chunk hashes, and completeness; startup and experiments reject stale or partial indexes. The candidate was generated against commit `0633c9128ea5` and tree `722644c84d2f` with working-tree dirty state `true`. Its manifest separately hashes every frozen evidence input, exact generator, and output; the earlier performance artefact retains its own execution commit and provenance digest. This local candidate is therefore not misrepresented as either the older experiment revision or a clean committed release.

B. Evaluation Protocol

The retrieval dataset has 50 source-derived cases across factual, methods, results, limitations, bibliographic, topic, comparison, synthesis, hard-negative, and unanswerable strata. One AI process created evidence-linked labels; a later shuffled, source-first pass re-inspected them without rankings. The fixed 30/20 split precedes tuning. On answerable queries, set recall is $|G_q \cap P_{q,k}|/|G_q|$; MRR and nDCG@10 complement it. 10000 query-bootstrap samples form percentile intervals, and paired randomization tests compare the tuned hybrid with baselines under Holm correction.

The QA run reuses 50 cases but intentionally evaluates the current untuned heuristic hybrid plus deterministic extractive answerer, not the best retrieval mode. Atomic claims and returned citations were checked for support, citation choice/completeness, answer-point coverage, and abstention. The finder comparison uses 28 synthetic profiles spanning skills, interests, time, project type, data, and difficulty constraints. It returns 84 items per arm. The evidence-only arm returns papers and passages; the full arm adds a separately labelled system suggestion. Topic evaluation uses 60 AI-silver papers (36 development and 24 test) and a 40-label vocabulary including an other-or-unknown label. Confidence intervals quantify case/profile sampling only; they do not model correlated AI-label error or constitute inter-rater reliability.

Candidate-specific protocols prevent favorable post hoc substitutions. Retrieval weights are selected only on development MRR from a recorded search space, then frozen for test. The QA answerer and retriever are named even though they are not the best-scoring combination. Recommendation contrasts resample complete profiles because three suggestions from one profile are correlated. Topic threshold selection uses development micro-F1, while held-out per-label support is retained so labels with no gold examples cannot inflate a macro narrative. Performance uses isolated paths and failures remain in denominators.

Figures 1 and 2 summarize the implementation. The React interface provides papers, search, Ask, Finder, topic/author, evaluation, and protected review routes. It renders freshness, provider, warnings, review state, citations, and evidence-only alternatives, including explicit empty, unavailable, and retry states. FastAPI keeps public reads/transient generation separate from authenticated mutations; request limits, exact origin/host checks, safe file/URL handling, redacted logs, and append-only attributed review events are enforced. Profile inputs are transient in evaluated calls.

Page-aware extraction distinguishes blank, scanned, and mixed pages. No corpus OCR was required, so OCR quality is supported only by engineering fixtures. Search returns the mode, score components, query expansion, pages, sections, and freshness. Runtime grounding values check locator presence and weak lexical overlap, not entailment. The answerer extracts bounded sentences. The finder first constructs source-backed baseline work, future work, and gap fields; only then does it add a labelled suggestion with MVP, stretch scope,

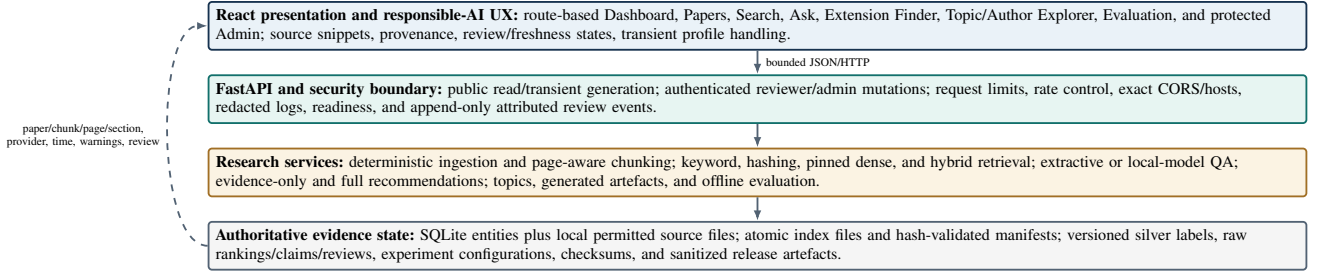


Fig. 1. Implemented architecture. Administrative mutations cross an authenticated boundary; evidence locators return through every layer.

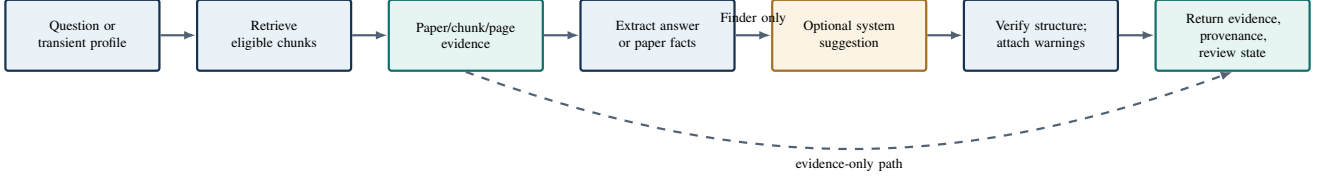


Fig. 2. Unified Ask/Finder evidence flow. Evidence-only output and system suggestions remain distinct.

TABLE II
HELD-OUT RETRIEVAL ESTIMATES (19 ANSWERABLE QUERIES).

Mode	R@3	R@10	MRR	nDCG@10
Keyword	0.939	1.000	0.947	0.958
Feature hashing	0.654	0.704	0.588	0.582
Learned dense	0.916	0.989	0.939	0.939
Heuristic hybrid	0.795	0.917	0.813	0.829
Dev-tuned hybrid	0.856	0.925	0.860	0.871

TABLE III
QA OUTCOMES OVER 50 CASES AND 400 CLAIMS.

Outcome	Estimate/count	95% CI
Supported / partial / unsupported	398/2/0	–
Strict supported-claim rate	0.995	[0.987, 1.000]
Citation correctness	0.625	[0.564, 0.691]
Citation presence	1.000	[1.000, 1.000]
Answer-point coverage	0.136	[0.067, 0.211]
Unanswerable abstention	0.000	[0.000, 0.000]

skills, data, risk, and evaluation. Optional Ollama was unreachable during evaluation, so no model tag, digest, response, or latency comparison was invented.

IV. RESULTS

A. RQ1: Corpus Integrity and Retrieval

All three authoritative indexes covered 719/719 eligible chunks; excluded/raw chunks could not enter evaluation. Author cleanup removed navigation artefacts and repaired punctuation aliases, while 13 possible same-person pairs remain unmerged pending external confirmation. Missing metadata stays empty.

Table II shows that keyword had the highest point estimates: Recall@10 1.000 (95% CI [1.000, 1.000]), MRR 0.947 ([0.868, 1.000]). Dense MRR was 0.939 ([0.842, 1.000]); tuned-hybrid MRR was 0.860 ([0.719, 0.974]). No corrected paired comparison established tuned-hybrid superiority, and equivalence was not tested. Removing the topic adjustment raised tuned Recall@10 to 0.978; removing all 6 heuristics recovered Recall@10 1.000 and MRR 0.947. Error coding found 2 intent/ranking, 1 extraction, and 1 corpus-absence signals. Every mode returned a result for the single held-out unanswerable query; the resulting 1.0 false-positive estimate is unstable but directionally consistent with QA abstention failure.

B. RQ2: Answers, Advice, Topics, and Review

Table III prevents a high “grounding” value from hiding distinct failures. All 400 checkable claims carried citations; AI review labelled 398 supported, 2 partial, and 0 unsupported. However, only 0.625 of claim–citation links were fully correct, expected answer-point coverage was 0.136, and none of 4 unanswerable cases abstained. Errors were usually supported but off-target extracts, evidence-selection misses, or omissions—not invented text. These labels are AI-reviewed silver judgments with no independent human adjudicator; two-pass consistency does not quantify their uncertainty.

Finder evidence-only relevance averaged 0.631 versus 0.655 for the full finder; the paired change 0.024 had CI [−0.024, 0.071], spanning zero. Source fidelity and fact/future-work/gap/suggestion separation passed all 84 full-arm items, yet all 84 feasibility ratings and 12 evaluation plans were partial. Thus schema compliance is not advisory utility. Sensitivity across 22 configurations found interest weight most disruptive.

On 24 topic test papers, controlled lexical precision was 0.652, recall 0.484, and F1 0.556; coverage was 0.875. Dense precision was 0.417, recall 0.694, F1 0.521, and coverage 0.958. Only 26 labels had held-out gold support and 14 had none, so vocabulary-wide claims would be misleading. The inspectable lexical method was retained for precision and governance, not universal superiority. Author links mean

TABLE IV
FAILURE-ORIENTED DENOMINATORS RETAINED FOR INTERPRETATION.

Component	Outcome	Count
QA citations	correct / partial / incorrect	250/142/8
QA answer points	full / partial / missing	11/37/33
QA returned citations	used in reviewed claims	148/250
Finder	feasibility rated partial	84/84
Finder	evaluation plan rated partial	12/84
Topics	labels with / without test support	26/14
Review	answers needing reprocessing	27

TABLE V
REPRESENTATIVE HELD-OUT LEXICAL TOPIC ERRORS.

Label	TP	FP	FN	Precision	Recall
Agriculture	2	1	0	0.667	1.000
Ai	0	0	4	0.000	0.000
Clustering	3	1	0	0.750	1.000
Energy	1	0	2	1.000	0.333
Networks	0	2	2	0.000	0.000
Rag	2	0	0	1.000	1.000

indexed-publication evidence only. Review appended 48 hash-chained AI events. Current recommendations and artifacts were regenerated, but 27 historical answers lacked sufficient stored scope and remain marked for reprocessing, not approved.

C. Cross-Component Failure Analysis

Table IV localizes what aggregate rates conceal. The 398/2/0 support distribution reflects an extractive answerer whose sentences usually came from some returned source, but citation review divided 400 links into 250 correct, 142 partial, and 8 incorrect links. Only 148 of 250 distinct returned citations supported a reviewed claim. Of 81 expected points, 11 were fully covered, 37 partially covered, and 33 absent. The error pattern therefore points to intent-aware retrieval and within-document evidence selection, not merely more fluent generation.

Recommendation failures occur at a different boundary. Every item supplied the required source/suggestion fields, but feasibility remained partial because profile constraints were matched lexically or heuristically rather than confirmed against data access, supervisor expectations, prerequisite mastery, or a calendar. Evaluation-plan partials clustered where an output named a metric without a defensible baseline, dataset split, or success threshold. A responsible next design should treat these fields as questions to resolve with a supervisor, not confidence-bearing facts.

Topic errors also resist one scalar summary. Held-out support covered only 26 of 40 labels; 14 labels had no positive test example. The lexical method missed broad implicit concepts but produced fewer speculative labels, while the dense prototype increased recall at the cost of precision. Per-label evidence and an `other/unknown` path are therefore part of the interface contract. Similarly, unresolved author pairs remain separate because merging identity records is a governance decision with consequences for attribution.

TABLE VI
SELECTED LOCAL STAGE MEDIANS (S) AND PEAK RSS (MiB).

Stage	Cold	Warm	RSS
Discovery	3.240	0.970	65.2
PDF extraction	15.466	12.874	112.6
Dense index	145.005	66.169	751.5
Hybrid retrieval	93.355	20.326	568.9
Offline answer	83.355	9.842	571.9
Rendered routes	12.496	12.463	458.2

Table V illustrates heterogeneous behavior behind micro-F1. Explicit terms supported perfect recall for the two `rag` positives, whereas the broader `ai` label missed all four positives and `networks` produced both misses and false positives. `Agriculture` and `clustering` were more recoverable; `energy` was precise when fired but missed two of three positives. The counts are too small for stable label rankings, but they identify which synonyms/rules require source inspection and which concepts may need a hierarchy rather than a flat vocabulary.

D. RQ3: Reproduction and Operational Boundary

The full local profile measured 17 stages with three process-cold and three warm repetitions at concurrency one: $17 \times 6 = 102$ samples, with 0 failures on an AMD Ryzen 7 5800X/16-logical-CPU WSL2 host. “Cold” means a fresh process; OS caches were not flushed, and this is neither load nor capacity evidence. The external harness reacquired 3 pinned CC BY JATS/XML papers and obtained 3/3 fixed lexical top-one matches; that establishes format compatibility and trivial discrimination only, not cross-domain quality or PDF ingestion.

Table VI selects representative stages from the full artifact; omitted stages and all repetitions remain in the hashed raw record. Network discovery dominates variability, dense indexing dominates local compute/memory, and warm/cold differences are workload-specific. No concurrency sweep, cache flush, or saturation test was conducted. The reproduction command rebuilds allowed derived state, runs evaluation/tests, compiles manuscripts, preflights PDFs, creates a restricted-content-free bundle, and emits checksums. Exact full-corpus reruns still require lawful access to the local PDFs and pinned dense model.

V. DISCUSSION, THREATS, AND ETHICS

The principal result is an evidence boundary, not a new ranking algorithm. Complete eligible indexing does not repair 36 unavailable/review records. Keyword outperformed hybrid point estimates, and ablations implicated intuitive topical heuristics; larger human judgments are required before method selection. High extractive support coexisted with poor citation targeting, completeness, and abstention, showing why locators and entailment cannot share one badge. Finder outputs cleanly separated source facts from suggestions, but their relevance difference was uncertain and every feasibility rating remained

partial. These negative results narrow the next experiment: calibrate answerability, improve within-paper evidence selection, require claim-to-citation verification, and evaluate candidate projects with students and supervisors rather than adding another generator. Human studies must test usefulness, novelty, workload, and trust.

A. Evidence-Driven Design Decisions

The retrieval evidence changes product defaults. Keyword search is the defensible default for this snapshot because it has the strongest held-out point estimates and simple failure semantics. Dense search remains valuable as an explicit alternative for paraphrase, but not as an automatic claim of semantic quality. The development-tuned hybrid remains experimental: its weights are reproducible and its score components visible, yet neither the paired analysis nor its ablations justify promoting it as superior. The interface therefore returns the active mode and diagnostic components and permits a user to switch modes. A future query router must earn its complexity on a new locked test set rather than on the cases that motivated it.

The QA evidence requires a multi-dimensional result state. A response can be source-derived but answer the wrong intent; cite every claim but choose only partially correct passages; cover one expected point while omitting others; or answer confidently when the corpus has no evidence. Consequently the public surface should show answerability, claim support, citation correctness/coverage, and answer completeness as separate review concepts. For a live answer these metrics may be unavailable, so the UI reports provenance and structural checks without projecting aggregate experiment scores onto an individual response. An abstention gate should first test corpus/query fit, then retrieval sufficiency, and finally claim support. If any stage fails, returning ranked evidence or an explicit unsupported state is safer than composing text.

The Finder needs an analogous decision boundary. Ranking a paper can be justified by passages, declared interests, and constraints; proposing a thesis adds claims about novelty, data access, workload, skills, and evaluation that the corpus cannot settle alone. The implemented record therefore distinguishes baseline work, paper-stated future work, inferred gap, and system suggestion. It gives the user an evidence-only mode, and the full mode returns assumptions and risks rather than a hidden suitability score. The observed all-partial feasibility result is useful: it says the next component should be a structured supervisor checklist and constraint-resolution workflow, not more assertive prose. No recommendation is labelled supervisor-approved until an authorized human records that state.

Administrative review is part of the evidence model rather than a content overwrite. Events identify actor type, prior and new states, rationale, time, target hash, and chain hash. AI actors can mark AI-reviewed or needs-reprocess states but cannot impersonate human approval. Historical answers that cannot be exactly regenerated stay visible with their limitation. Likewise, ambiguous author identities remain candidates

instead of automatic merges. These choices sacrifice apparent completeness to preserve attribution and make later correction auditable.

The external smoke test and performance profile answer deployment questions only narrowly. The former shows that three licensed XML documents can enter the chunk contract and be separated by obvious lexical queries; it does not validate another discipline, language, or PDF path. The latter identifies expensive stages for a single local operator but offers no arrival-rate, tail-latency-under-load, or capacity evidence. Production planning must add representative traffic, concurrent users, failure injection, monitoring, and recovery tests without reusing these single-user medians as service-level objectives.

Construct validity is limited by AI-silver relevance/claim labels and synthetic-profile proxy ratings. The same AI process created and re-inspected labels, so correlated error remains; intervals exclude label uncertainty. Internal validity is limited by small held-out sets, a single retrieval unanswerable, post-selection descriptive ablations, and a deterministic answerer unlike all RAG systems. External validity is limited to one laboratory snapshot, English extraction, local hardware, and a three-document format smoke test. Reproducibility is bounded by lawful PDF/model access, network discovery, OS caches, and the explicitly dirty candidate state.

TTLAB authorized project and corpus use. No participants were recruited and no ethics approval or exemption is claimed. Authorization does not transfer redistribution rights: PDFs, live database, derived text, secrets, model caches, and private prompts are excluded from the sanitized release. Public deployment additionally requires authenticated review, controlled origins, rate limits, monitoring, correction paths, backups, and accessibility validation. Datasheet, model-card, and risk-management practices inform disclosure [19]–[21]. Codex/LLM assistance covered code, silver-label preparation, analysis, and editing; it was not a human assessor. The current Tectonic PDF is intentionally not claimed as tagged PDF or PDF/UA; figure descriptions remain in source for a capable venue toolchain.

VI. AVAILABILITY AND CONCLUSION

The sanitized bundle contains code, permitted metadata, sanitized experimental records, configurations, figure sources, manifests, hashes, and reproduction instructions. It excludes restricted content and does not claim to be the final publisher package. For this snapshot, full-paper search and advice are inspectable: dense MRR 0.939 and strict claim support 0.995 coexist with a negative hybrid result, citation correctness 0.625, coverage 0.136, failed abstention, and uncertain recommendation benefit. The artefact is therefore an evaluated research-intelligence prototype, not a validated academic advisor. Human utility, identity decisions, rights, and venue compliance remain external validations.

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